

The Architecture of My Large Language Model

and an open plan to beat SmolLM

A Technical Note by **Vuk Rosić**

A technical note by **Vuk Rosić** · Open Superintelligence Lab. Open code: github.com/vukrosic/universe-lm. All figures are original schematics.

Looking for compute & collaborators. I am building this language model fully in the open and racing to beat SmolLM-135M. If you have GPUs to run open-source experiments — or want to help build it — the request is in §6, or reach me in **Zhongguancun, Beijing** for an in-person chat.

Abstract

This note describes the architecture of universe-lm, a small language model I train completely in the open, and the plan around it. The model is a standard decoder-only transformer — grouped-query attention, rotary positions, RMSNorm and a squared-ReLU feed-forward — with four small, cheap, zero-initialized additions to attention that are the actual research: value embeddings, a per-head query gain, a tuned rotary base, and sliding-window attention. The first half explains how the model works; the second half explains why it is worth compute: everything is published the same day it runs (no multi-month paper embargo), the goal is to beat SmolLM-135M on its public eval suite at matched tokens, and I am asking for roughly 1,200–1,500 A100-hours, tiered, to do it. Every experiment is open-source, so a provider runs the exact public code.

The model: a decoder-only transformer

Tokens in, next-token probabilities out. A standard modern stack — the universe-lm-specific ideas live inside each block (next figure).

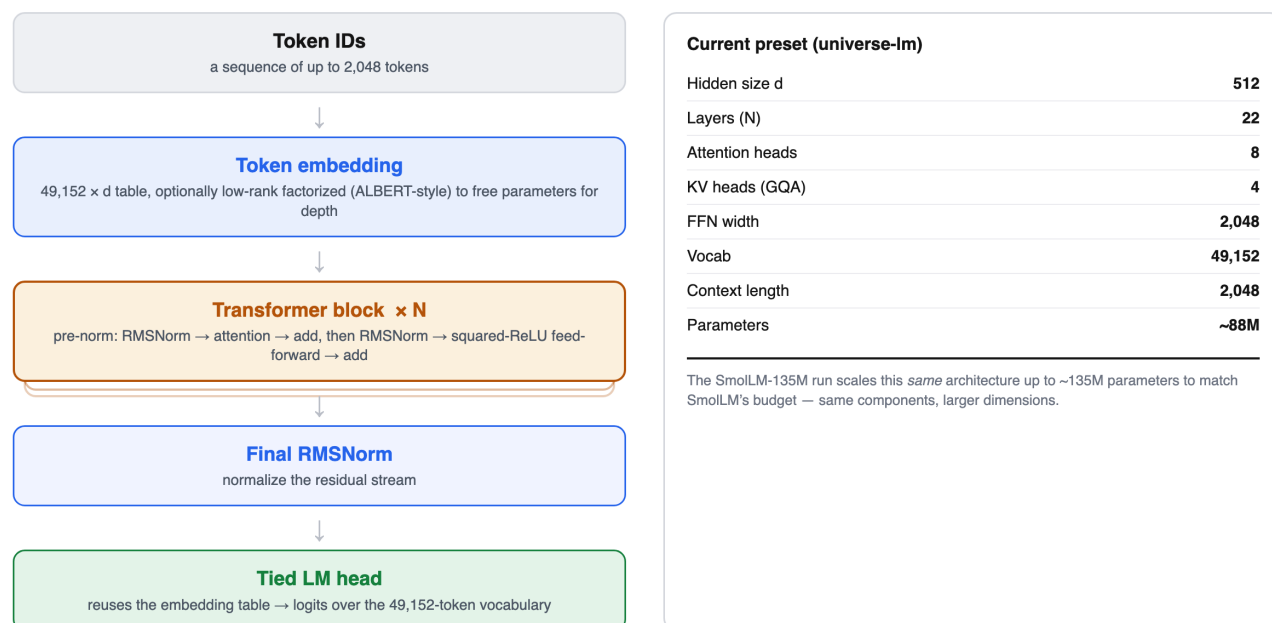


Figure 1. The model, end to end (original schematic). Token IDs become embeddings, flow through N identical pre-norm transformer blocks, and exit through a tied head as next-token probabilities over the 49,152-token

vocabulary. The current preset is ~88M parameters.

1 The architecture: a decoder-only transformer

universe-lm reads a sequence of tokens and predicts the next one. Tokens are looked up in an embedding table (optionally low-rank factorized, ALBERT-style, to spend fewer parameters on the vocabulary and more on depth), passed through a stack of pre-norm transformer blocks, normalized once more, and projected back to vocabulary logits through a head that reuses the embedding table (weight tying). The components — grouped-query attention, rotary position encoding, RMSNorm and a squared-ReLU feed-forward — are the modern standard. The current preset is ~88M parameters; the SmolLM-135M run scales this same architecture up to match SmolLM’s budget.

2 Inside a block, and the four ideas

Every block has two pre-norm sub-layers: grouped-query attention (8 query heads sharing 4 key/value heads, with rotary positions), then a squared-ReLU feed-forward. The research lives in four small additions to attention, each *zero-initialized* so the model begins exactly as the plain baseline and only earns a change during training if it actually helps: **value embeddings** (push token identity straight into attention’s value vector), a **per-head query gain** (a learnable per-head temperature on attention sharpness), a tuned **rotary base** (reshapes how distance is encoded), and **sliding-window attention** (attend to a local window, cheaper and better at small scale).

Inside one block — and the four ideas being tested

Each block has two pre-norm sub-layers. The universe-lm ideas hook into the attention sub-layer, and each is zero-initialized so the model starts exactly as the baseline and earns the change during training.

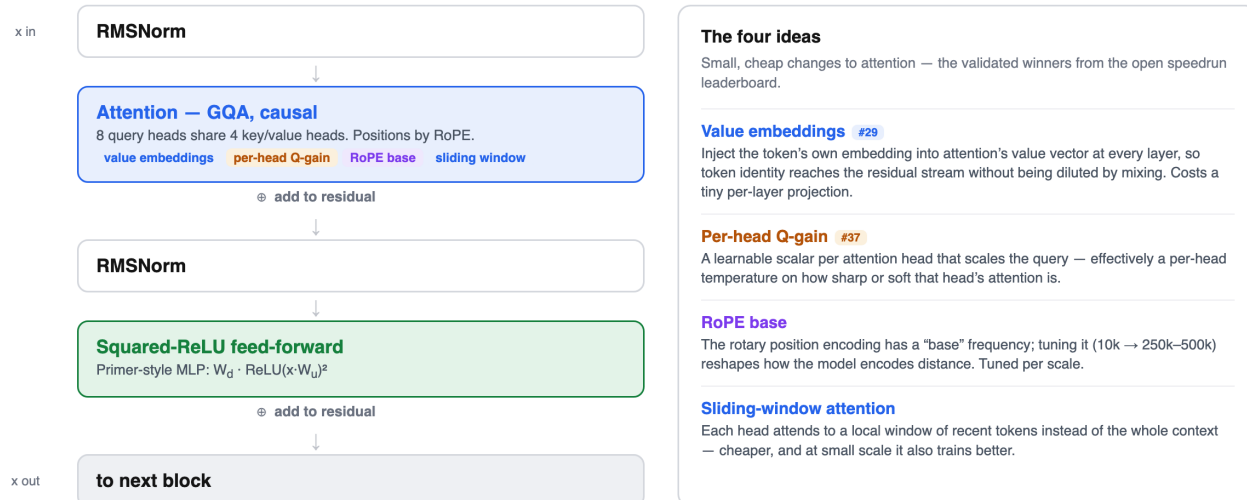


Figure 2. One block and the four ideas (original schematic). The ideas all hook into the attention sub-layer; zero-init means each starts as a no-op and is kept only if training shows it helps.

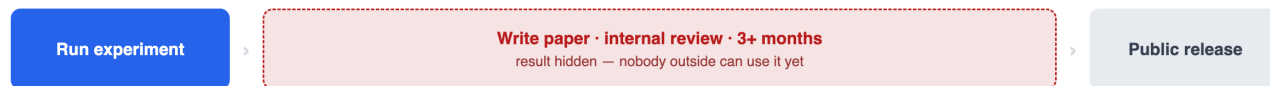
3 Why this is worth compute: open in time, not just open in license

The case for funding this is about timing. OLMo, EleutherAI and HuggingFace’s SmolLM are genuinely open, but they release on publication — a finished result is held for months while a paper is written and reviewed, and only then can outsiders use it. universe-lm publishes code, data, configs, logs and results the day each run finishes, so the same experiments produce more total scientific value with no embargo.

Daily-open vs. publish-then-open

Even fully-open labs release on publication. The result exists for months before anyone outside can build on it.

Typical open lab — OLMo / EleutherAI / SmolLM style



The science is real on day 0, but the world only gets it after the embargo.

universe-lm — daily-open



Code, data, configs, logs and results pushed to GitHub the same day each run finishes — zero embargo.

Figure 3. Daily-open vs. publish-then-open (original schematic). Most open labs (top) sit behind a multi-month embargo; universe-lm (bottom) pushes everything to GitHub the day each run finishes.

4 The goal and the scale: beat SmolLM-135M, iterate overnight

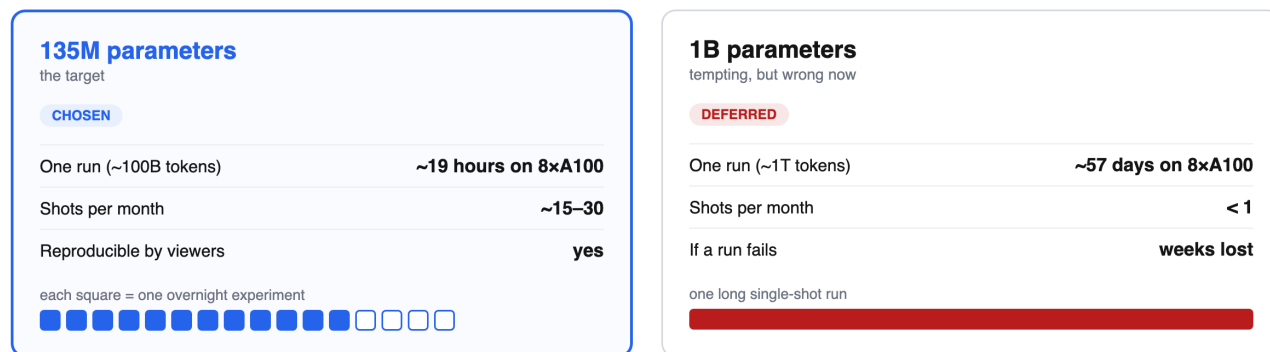
The only claim worth making is a matched-token, downstream-benchmark result anyone can reproduce. Training compute scales as:

$$C \approx 6ND$$

where N is parameters and D is training tokens. At 135M a full run is an overnight job — many shots a month; a 1B run is a roughly two-month single shot that kills iteration. The data recipe proven at 135M transfers upward, so small first is the fast path to scale, not a detour.

Why 135M, not 1B — iteration speed is the science

Training cost $\approx 6 \times \text{params} \times \text{tokens}$. On one A100 (~150 TFLOP/s realized). The smaller target gives many shots; the larger one gives a single slow one.



Win condition either way: beat SmolLM-135M on its public eval suite (HellaSwag, ARC, PIQA...) at a matched token budget. The data recipe proven at 135M transfers up later — so small first is the fast path to big, not a detour.

Figure 4. Why 135M, not 1B (original schematic). 135M finishes overnight on one 8-GPU node (~15–30 runs/month); 1B is a single ~2-month shot. Same win condition either way.

5 The open data pipeline: the real lever

Small models are won on data quality, not architecture — that was SmolLM's own finding. The pipeline filters open web text for educational quality, deduplicates, and optionally adds synthetic textbook-style data; the code is versioned in the repo and the processed dataset is published on the HuggingFace Hub, so the full path from raw text to a checkpoint is reproducible.

Open data pipeline — raw text to trained weights, all reproducible

Pipeline *code* lives in the public repo; the processed *dataset* is published on the HuggingFace Hub. Anyone can rerun every step.

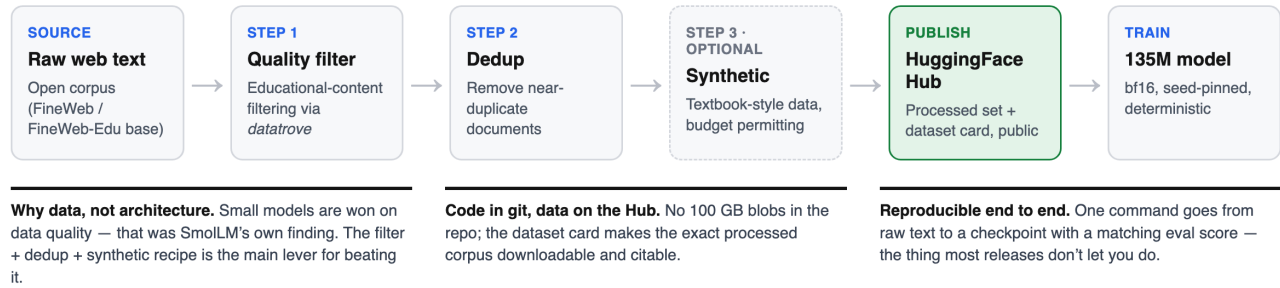


Figure 5. The open data pipeline (original schematic). Code in git, data on the Hub — the end-to-end reproducibility most releases do not provide.

6 The experiment program and the compute request

A 135M model fits on a single GPU, so training needs no tensor or pipeline parallelism — it is plain data parallelism (DDP / FSDP) on one multi-GPU node. The program is ~10–15 public runs in four phases. The request is roughly 1,200–1,500 A100-hours, tiered: about 200 hours to validate the pipeline and the first real result, then scaling as the numbers hold, so the first commitment is low-risk. Every experiment is open-source on GitHub, so a provider runs the exact public code. I am based in Zhongguancun, Beijing and open to meeting in person. *Scope.* The figures are schematics of a proposed project; compute-hour estimates use 6ND at ~150 TFLOP/s realized per A100 and will be refined against the actual cluster and tokenizer.

Experiment program and the compute ask

~10–15 runs, every one open-source on GitHub. The provider runs the public code — nothing hidden, nothing proprietary.

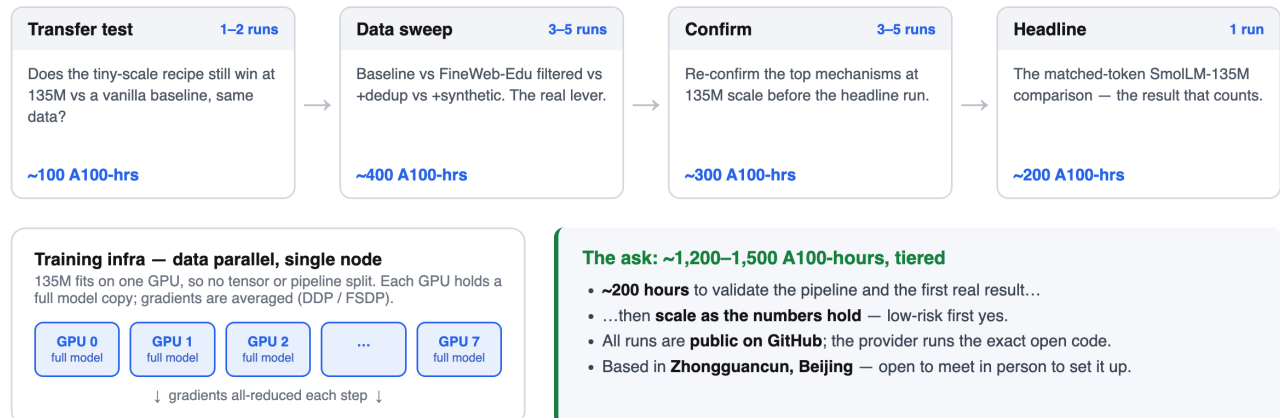


Figure 6. The experiment program and compute request (original schematic). Four phases, every run open on GitHub; single-node data-parallel training; ~1,200–1,500 A100-hours, tiered.

Architecture	Decoder-only transformer: GQA + RoPE + RMSNorm + squared-ReLU FFN, tied head.
The four ideas	Value embeddings, per-head Q-gain, tuned RoPE base, sliding-window attention (all zero-init).
Goal	Beat SmolLM-135M on its public eval suite at a matched token budget, fully reproducible.
Openness	Code, data, configs, logs and results published daily — no publication embargo.
Scale & training	135M params; data-parallel (DDP/FSDP), single node; overnight runs.
Compute ask	~1,200–1,500 A100-hours, tiered. Zhongguancun, Beijing — open to meet.

Table 1. At a glance.

References

- [1] V. Rosić. *universe-lm: an open, daily-published small language model*. Source code, configs and logs, vukrosic/universe-lm repository.
- [2] HuggingFace. *SmolLM and SmolLM2: small language models trained on curated open data*. Models, datasets and documentation.
- [3] HuggingFace. *FineWeb and FineWeb-Edu: open web datasets, and the datatrove processing library*. Datasets and documentation.

Author’s note. This is a technical note and proposal by Vuk Rosić (Open Superintelligence Lab), dated 28 June 2026. It describes the universe-lm architecture and planned open work, not a deployed result; the figures are original schematics and the compute-hour figures are estimates. Named models, datasets and libraries (SmolLM, FineWeb, datatrove) are the work of their respective authors. Any errors are the author’s own.